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## Interferometric detection and quantification system and methods of use in healthcare

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### Abstract

A point of care analyte detection and quantification system for healthcare application is provided. Related methods are also provided.


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## References Cited

**U.S. PATENT DOCUMENTS**

| Patent No. | Issued Date | Patentee Name | U.S. Cl. | CPC |
|---|---|---|---|---|
| 4940328 | 12/1989 | Hartman | N/A | N/A |
| 5082629 | 12/1991 | Burgess, Jr. et al. | N/A | N/A |
| 5262842 | 12/1992 | Gauglitz et al. | N/A | N/A |
| 5279791 | 12/1993 | Aldrich et al. | N/A | N/A |
| 5320808 | 12/1993 | Holen et al. | N/A | N/A |
| 5405510 | 12/1994 | Betts et al. | N/A | N/A |
| 5504011 | 12/1995 | Gavin et al. | N/A | N/A |
| 5603351 | 12/1996 | Cherukuri et al. | N/A | N/A |
| 5623561 | 12/1996 | Hartman | N/A | N/A |
| 5731212 | 12/1997 | Gavin et al. | N/A | N/A |
| 5787885 | 12/1997 | Lemelson | N/A | N/A |
| 6078705 | 12/1999 | Neuschäfer et al. | N/A | N/A |
| 6192168 | 12/2000 | Feldstein et al. | N/A | N/A |
| 6198869 | 12/2000 | Kraus et al. | N/A | N/A |
| 6242267 | 12/2000 | Herron et al. | N/A | N/A |
| 6312961 | 12/2000 | Voirin et al. | N/A | N/A |
| 6326612 | 12/2000 | Elkind et al. | N/A | N/A |
| 6469785 | 12/2001 | DuVeneck et al. | N/A | N/A |
| 6483096 | 12/2001 | Kunz et al. | N/A | N/A |
| 7184626 | 12/2006 | Gunn, III et al. | N/A | N/A |
| 7630063 | 12/2008 | Padmanabhan et al. | N/A | N/A |
| 8502985 | 12/2012 | Weinberger et al. | 356/450 | G01B 9/02 |
| 8940247 | 12/2014 | Uehata et al. | N/A | N/A |
| 9739714 | 12/2016 | Moll et al. | N/A | N/A |
| 9936916 | 12/2017 | Sahin | N/A | A61B 5/00 |
| 10564373 | 12/2019 | Ogden et al. | N/A | N/A |
| 11740177 | 12/2022 | Levin et al. | N/A | N/A |
| 11747276 | 12/2022 | Levin et al. | 356/498 | G01N 21/45 |
| 11994504 | 12/2023 | Levin et al. | N/A | N/A |
| 12146829 | 12/2023 | Levin et al. | 356/477 | G01N 21/45 |

| | | | | |
|---|---|---|---|---|
| 2002/0098122 | 12/2001 | Singh et al. | N/A | N/A |
| 2002/0159050 | 12/2001 | Sharma et al. | N/A | N/A |
| 2002/0189945 | 12/2001 | Ruggiero | N/A | N/A |
| 2003/0186228 | 12/2002 | McDevitt et al. | N/A | N/A |
| 2004/0029259 | 12/2003 | McDevitt et al. | N/A | N/A |
| 2004/0202400 | 12/2003 | Kochergin et al. | N/A | N/A |
| 2004/0248318 | 12/2003 | Weinberger et al. | N/A | N/A |
| 2005/0088648 | 12/2004 | Grace et al. | N/A | N/A |
| 2005/0105077 | 12/2004 | Padmanabhan et al. | N/A | N/A |
| 2005/0135723 | 12/2004 | Carr et al. | N/A | N/A |
| 2005/0136551 | 12/2004 | Mpock | N/A | N/A |
| 2005/0157304 | 12/2004 | Xiao et al. | N/A | N/A |
| 2005/0185178 | 12/2004 | Gardner, Jr. et al. | N/A | N/A |
| 2006/0216200 | 12/2005 | Nagatomo et al. | N/A | N/A |
| 2006/0292039 | 12/2005 | Iida | N/A | N/A |
| 2006/0292040 | 12/2005 | Wickstead et al. | N/A | N/A |
| 2007/0146718 | 12/2006 | Takase et al. | N/A | N/A |
| 2008/0286858 | 12/2007 | Cracauer et al. | N/A | N/A |
| 2009/0316158 | 12/2008 | Ash, III | N/A | N/A |
| 2010/0227386 | 12/2009 | Neuzil et al. | N/A | N/A |
| 2010/0271634 | 12/2009 | Dominguez Horna et al. | N/A | N/A |
| 2011/0253224 | 12/2010 | Linder et al. | N/A | N/A |
| 2011/0256551 | 12/2010 | Linder et al. | N/A | N/A |
| 2012/0071342 | 12/2011 | Lochhead et al. | N/A | N/A |
| 2012/0088230 | 12/2011 | Givens et al. | N/A | N/A |
| 2012/0115214 | 12/2011 | Battrell et al. | N/A | N/A |
| 2012/0213669 | 12/2011 | Kasai | 422/69 | G01N 33/54373 |
| 2012/0214707 | 12/2011 | Ymeti et al. | 506/9 | C40B 30/04 |
| 2013/0004952 | 12/2012 | Castagna et al. | N/A | N/A |
| 2013/0004954 | 12/2012 | Bianchessi et al. | N/A | N/A |
| 2013/0196343 | 12/2012 | Mastrangelo et al. | N/A | N/A |
| 2013/0273528 | 12/2012 | Ehrenkranz | N/A | N/A |
| 2013/0330232 | 12/2012 | Pruessner et al. | N/A | N/A |
| 2013/0330245 | 12/2012 | Duncan et al. | N/A | N/A |
| 2014/0038222 | 12/2013 | Alt et al. | N/A | N/A |
| 2014/0065720 | 12/2013 | Ja | N/A | N/A |
| 2014/0080729 | 12/2013 | Grego et al. | N/A | N/A |
| 2014/0335527 | 12/2013 | Goel | N/A | N/A |
| 2015/0185159 | 12/2014 | Morita et al. | N/A | N/A |
| 2015/0244852 | 12/2014 | Erickson et al. | N/A | N/A |
| 2015/0293021 | 12/2014 | Finkelstein et al. | N/A | N/A |
| 2015/0301167 | 12/2014 | Sentelle et al. | N/A | N/A |
| 2016/0175840 | 12/2015 | Ingber et al. | N/A | N/A |
| 2016/0282184 | 12/2015 | Khalil et al. | N/A | N/A |
| 2016/0356708 | 12/2015 | Bennett et al. | N/A | N/A |
| 2017/0189906 | 12/2016 | Moll et al. | N/A | N/A |
| 2017/0203295 | 12/2016 | Bau-Madsen et al. | N/A | N/A |
| 2017/0227537 | 12/2016 | Guo et al. | N/A | N/A |

| | | | | |
|---|---|---|---|---|
| 2017/0292908 | 12/2016 | Wilk et al. | N/A | N/A |
| 2018/0117587 | 12/2017 | Lemoine et al. | N/A | N/A |
| 2018/0164242 | 12/2017 | Virtanen et al. | N/A | N/A |
| 2018/0304260 | 12/2017 | Thomas et al. | N/A | N/A |
| 2019/0056304 | 12/2018 | Gershtein | N/A | N/A |
| 2019/0187162 | 12/2018 | Shastry et al. | N/A | N/A |
| 2020/0003746 | 12/2019 | LeFiles et al. | N/A | N/A |
| 2020/0166453 | 12/2019 | Lendl et al. | N/A | N/A |
| 2020/0225145 | 12/2019 | Quidant et al. | N/A | N/A |
| 2020/0306757 | 12/2019 | Lee | N/A | N/A |
| 2020/0376494 | 12/2019 | DeJohn et al. | N/A | N/A |
| 2020/0400883 | 12/2019 | Laplatine et al. | N/A | N/A |
| 2021/0102901 | 12/2020 | Budnicki et al. | N/A | N/A |
| 2021/0293716 | 12/2020 | Carothers | N/A | N/A |
| 2021/0298651 | 12/2020 | LeFiles et al. | N/A | N/A |
| 2021/0301317 | 12/2020 | LeFiles et al. | N/A | N/A |
| 2021/0349018 | 12/2020 | Venkatarayalu et al. | N/A | N/A |
| 2022/0090908 | 12/2021 | Levin et al. | N/A | N/A |
| 2022/0091029 | 12/2021 | Levin et al. | N/A | N/A |
| 2022/0091030 | 12/2021 | Levin et al. | N/A | N/A |
| 2022/0091031 | 12/2021 | Levin et al. | N/A | N/A |
| 2022/0091086 | 12/2021 | Levin et al. | N/A | N/A |
| 2022/0091114 | 12/2021 | Levin et al. | N/A | N/A |
| 2022/0091119 | 12/2021 | Xu | N/A | N/A |
| 2022/0221438 | 12/2021 | LeFiles et al. | N/A | N/A |
| 2022/0229053 | 12/2021 | Levin et al. | N/A | N/A |
| 2022/0343271 | 12/2021 | Levin et al. | N/A | N/A |
| 2023/0025577 | 12/2022 | Scullion et al. | N/A | N/A |
| 2023/0194427 | 12/2022 | Alexandros et al. | N/A | N/A |
| 2023/0358674 | 12/2022 | Levin et al. | N/A | N/A |
| 2024/0027341 | 12/2023 | Levin et al. | N/A | N/A |

**FOREIGN PATENT DOCUMENTS**

| **Patent No.** | **Application Date** | **Country** | **CPC** |
|---|---|---|---|
| 2009115847 | 12/2008 | WO | N/A |
| 2012033466 | 12/2011 | WO | N/A |

**OTHER PUBLICATIONS**

U.S. Appl. No. 17/448,082, filed Sep. 20, 2020, Levin et al. cited by examiner
U.S. Appl. No. 17/448,075, filed Sep. 20, 2021, Levin et al. cited by examiner
Ymeti et al. ‘Fast, Ultrasensitive Virus Detection Using a Young Interferometer Sensor’, NANO Letters vol. 7 No. 2 pp. 394-397 2007 (Year: 2007). cited by examiner
Ymeti et al. ‘An ultrasensitive Young handheld Interferometer Sensor for rapid Virus Detection’, Future Drugs, pp. 447-453, 2007 (Year: 2007). cited by examiner
International Preliminary Report on Patentability dated Mar. 30, 2023 for PCT/US2021/071510. cited by applicant
Office Action in U.S. Appl. No. 17/448,082, dated Aug. 7, 2024. cited by applicant
Ymeti, A et al, Applied Optics 2003, 42, 5649-5660. (Year: 2003). cited by applicant
Sepulveda, B. et al, Journal of Optics A: pure and Applied Optics 2006, 8, S561-S566. (Year:

2006). cited by applicant
Kujawinska, M. et al, SPIE 2006, 6293, paper 62930T, 7 pages. (Year: 2006). cited by applicant
Ymeti, A et al, Nano Letters 2007, 7, 394-397. (Year: 2007). cited by applicant
Ymeti, A et al, Expert Review of Medical Devices 2007, 4, 447-454. (Year: 2007). cited by applicant
Xu, J. et al, Analytical and Bioanalytical Chemistry 2007, 389, 1193-1199. (Year: 2007). cited by applicant
Lechuga, L. M. et al, SPIE, 2009, 7220, paper 72200L, 8 pages. (Year: 2009). cited by applicant
Crespi, A et al, Lab on a Chip 2010, 10, 1167-1173. (Year: 2010). cited by applicant
Kozma, P. et al, Sensors and Actuators B 2011, 155, 446-450. (Year: 2011). cited by applicant
Zinoviev, K. E. et al, Journal of Lightwave Technology 2011, 29, 1926-1930. (Year: 2011). cited by applicant
Gonzalez-Guerrero, A B. et al, Procedia Engineering 2011, 25 71-75. (Year: 2011). cited by applicant
Dante, S. et al, Optica Pura Y Aplicada 2012, 45, 87-95. (Year: 2012). cited by applicant
Duval, D. et al, Lab on a Chip 2012, 12, 1987-1994. (Year: 2012). cited by applicant
Kozma, P. et al, Biosensors and Bioelectronics 2014, 58, 287-307. (Year: 2014). cited by applicant
Sarkar, D. et al, Biomedical Microdevices 2014, 16, 509-520. (Year: 2014). cited by applicant
Liu, Q. et al, Biosensors and Bioelectronics 201571, 365-372. (Year: 2015). cited by applicant
Ramirez, J.C. et al, Optics Express 2015, 23, paper 11985, 10 pages. (Year: 2015). cited by applicant
Aikio, S. et al, Journal of the European Optical Society—Rapid Publications 2015, 10, paper 15053, 7 pages. (Year: 2015). cited by applicant
Toccafondo, V. et al, Photonics Research 2016, 4, 57-60. (Year: 2016). cited by applicant
Alkio, S. et al, RSC Advances 2016, 6, 50414-50422. (Year: 2016). cited by applicant
Nabok, A et al, Sensors and Actuators B 2017, 247, 975-980. (Year: 2017). cited by applicant
Ghosh, S. et al, Journal of Lightwave Technology 2017, 35, 3003-3011. (Year: 2017). cited by applicant
Xhoxhi, M. et al, SPIE 2017, 10247, paper 102470F, 15 pages. (Year: 2017). cited by applicant
Al-Jawdah, A et al, Toxins 2018, 10, paper 272, 6 pages. (Year: 2018). cited by applicant
Zhou, C. et al, Optics Express 2018, 26, paper 24372, 12 pages. (Year: 2018). cited by applicant
Misiakos, K. et al, ACS Photonics 2019, 6, 1694-1705. (Year: 2019). cited by applicant
Extended European Search Report for European Patent Application No. 21870464.1, dated Sep. 11, 2024. cited by applicant
International Search Report and Written Opinion for PCT/US2021/071510, dated Dec. 29, 2021. cited by applicant
Office Action from U.S. Appl. No. 17/448,075 dated Oct. 31, 2024. cited by applicant
Office Action for U.S. Appl. No. 18/651,942 dated Dec. 23, 2024. cited by applicant

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application claims priority to U.S. Provisional Application No. 63/147,424 filed Feb. 9, 2021 and U.S. Provisional Application

No. 63/080,199 filed Sep. 18, 2020, the contents of which are each incorporated herein in their entirety.

## BACKGROUND

(1) Pathological and chemical contamination is a problem for all industries, including the healthcare industry, and for the general public alike. With an increase in understanding of global pandemics, public awareness of the presence of pathogens or harmful chemicals in, on or around the body have become grave concerns. There is also more public awareness of beneficial microbes in the body and environment. There also exists a need for high throughput, efficient in vitro diagnostic systems that can provide medical professionals and members of the public with information pertaining to qualitative and quantitative data for a variety of pathogens or chemical contaminants in a single test sample.

## SUMMARY

(2) A portable interferometric system for detection and quantification of analyte within a healthcare test sample composition is provided. The system includes an optical assembly unit, the optical assembly unit comprising a light unit and a detector unit each adapted to fit within a portable housing unit; and a cartridge system adapted to be inserted in the housing and removed after one or more uses, the cartridge system comprising an interferometric chip and a flow cell wafer. The interferometric chip includes one or more waveguide channels having a sensing layer thereon, the sensing layer adapted to bind or otherwise be selectively disturbed by one or more analytes within the healthcare test sample composition.

(3) According to one embodiment, the portable housing is sized and shaped to fit in a user's hand. According to one embodiment, the portable interferometric system further includes at least one display unit. According to one embodiment, the portable interferometric system further includes an external camera, the external camera adapted to capture a photo or video. According to one embodiment, the portable interferometric system includes an alignment means for aligning the cartridge system within a cartridge recess in the interferometric system. According to one embodiment, the sensing layer includes one or more antigens, antibodies, DNA microarrays, polypeptides, nucleic acids, carbohydrates, lipids, or molecularly imprinted polymers, or immunoglobulins suitable for binding one or more analytes within a healthcare test sample composition. According to one embodiment, the portable interferometric system is configured to analyze the light signals from two or more waveguide channels to detect the presence of an analyte that individual waveguides could not have detected alone. According to one embodiment, the one or more waveguide channels each comprises a different sensitive layer to allow the system to detect different analytes on each waveguide channel. According to one embodiment, the sensitive layer is configured to bind one or more chemical, antibody, virus antigen, virus protein, bacteria, fungi, pathogen, RNA, mRNA, plant growth regulator, metal or any combination thereof. According to one embodiment, the portable interferometric system exhibits an analyte detection limit down to about 1.0 picogram/L. According to one embodiment, the portable interferometric system exhibits an analyte detection limit down to about 1000 pfu/ml. According to one embodiment, the portable interferometric system exhibits sensitivity to at least 2 pixels per diffraction line pair. According to one embodiment, the portable interferometric system further includes a location means adapted to determine the physical location of the system. According to one embodiment, the analyte is one or more of a fungicide, herbicide, insecticide, fungus, bacterium, or microbe.

(4) A method of detecting and quantifying the level of analyte in a healthcare test sample composition is provided. The method includes the steps of: collecting a healthcare target sample containing one or more analytes; optionally entering an identification associated with the target sample; introducing the healthcare target sample to an interferometric system as provided herein; optionally, mixing the target sample with a buffer solution to form a healthcare test sample composition; initiating waveguide interferometry on the test sample composition; processing any

data resulting from the waveguide interferometry; and optionally, transmitting any data resulting from the waveguide interferometry. According to one embodiment, the step of transmitting data includes wirelessly transmitting analyte detection and quantification data to a mobile device or server. According to one embodiment, the method further includes the step of displaying data related to the presence of analyte in the test sample composition on the display unit. According to one embodiment, the healthcare target sample is taken from a bodily fluid or gaseous emission of the body. According to one embodiment, the bodily fluid is blood, urine, mucus or saliva. According to one embodiment, the method of detecting and quantifying the level of analyte in a healthcare test sample composition further includes the step of initiating a cleaning or remedial countermeasure against an analyte. According to one embodiment, the healthcare target sample is taken from water, soil, air, exhaled breath, skin, hair, or a bodily fluid or gaseous emission of the body. According to one embodiment, the healthcare target sample is in the form of, dissolved in, or suspended in a liquid or a gas. According to one embodiment, the data resulting from the waveguide interferometry is provided at or under 30 minutes.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** illustrates a perspective view of one embodiment of a handheld interferometric system as provided herein.

(2) FIG. **2**A illustrates a front view of one embodiment of a handheld interferometric system as provided herein.

(3) FIG. **2**B illustrates a rear view of one embodiment of a handheld interferometric system as provided herein.

(4) FIG. **3**A illustrates a cross-sectional view of an interferometric chip that may be integrated into a cartridge system as provided herein.

(5) FIG. **3**B illustrates a bottom view of a flow cell wafer having a serpentine shaped detection microchannel.

(6) FIG. **3**C illustrates a top view of a chip illustrating the movement of a light signal through the chip.

(7) FIG. **4** illustrates a side view of one embodiment of an optical assembly typically found in the handheld interferometric system of FIG. **1**.

(8) FIG. **5**A illustrates a cross-sectional view of the optical assembly of FIG. **4**.

(9) FIG. **5**B illustrates an alignment means according to one embodiment.

(10) FIG. **5**C illustrates an embodiment of a top view of the optical assembly and alignment means.

(11) FIG. **6** illustrates the cross-sectional view of the optical assembly of FIG. **5**A with one embodiment of a cartridge system inserted in the optical assembly.

(12) FIG. **7** illustrates a top view of the optical assembly of FIG. **5**A with one embodiment of a cartridge system inserted in the optical assembly.

(13) FIG. **8**A illustrates a view of the top surface of one embodiment of a single-use cartridge system.

(14) FIG. **8**B illustrates a view of the bottom surface of one embodiment of a single-use cartridge system.

(15) FIG. **8**C illustrates a view of the back surface of one embodiment of a single-use cartridge system.

(16) FIG. **8**D illustrates a view of the front surface of one embodiment of a single-use cartridge system.

(17) FIG. **8**E illustrates a view of one side surface of one embodiment of a single-use cartridge system.

(18) FIG. **8**F illustrates a cross-section view (looking downward) of a single-use cartridge system along the horizontal line of FIG. **8**E.
(19) FIG. **9**A illustrates a view of the top surface of one embodiment of a multi-use cartridge system.
(20) FIG. **9**B illustrates a view of the bottom surface of one embodiment of a multi-use cartridge system.
(21) FIG. **9**C illustrates a view of the back surface of one embodiment of a multi-use cartridge system.
(22) FIG. **9**D illustrates a view of the front surface of one embodiment of a multi-use cartridge system.
(23) FIG. **9**E illustrates a side surface view of one embodiment of a multi-use cartridge system.
(24) FIG. **9**F illustrates a cross-section view (looking downward) of one embodiment of a multi-use cartridge system along the horizontal line of FIG. **9**E.
(25) FIG. **10** illustrates a perspective view of an alternative single-use cartridge system.
(26) FIG. **11** illustrates a method of detecting and quantifying the level of analyte in a healthcare test sample composition.
(27) FIG. **12**A illustrates a quantification and monitoring system for analytes within an aqueous target sample from a rinse sink.
(28) FIG. **12**B illustrates a quantification and monitoring system for analytes within an aqueous target sample from a suction line.

DETAILED DESCRIPTION

(29) One or more aspects and embodiments may be incorporated in a different embodiment although not specifically described. That is, all aspects and embodiments can be combined in any way or combination. When referring to the compounds disclosed herein, the following terms have the following meanings unless indicated otherwise. The following definitions are meant to clarify, but not limit, the terms defined. If a particular term used herein is not specifically defined, such term should not be considered indefinite. Rather, terms are used within their accepted meanings.

Definitions

(30) As used herein, the term “portable” refers to the capability of the interferometric systems described herein to be transported or otherwise carried to a target sample location for use according the methods provided herein.
(31) As used herein, the term “analyte” refers to a substance that is detected, identified, measured or any combination thereof by the systems provided herein. The analyte includes any solid, liquid, or gas affecting (positively or negatively) an environment of interest. The analyte can be beneficial or deleterious. The analyte includes, but is not limited to, chemicals as well as bacteria and other pathogenic microorganisms that may negatively or positively impact health. The analyte includes, but is not limited to microbes (beneficial or pathogenic that may be dead or alive), biomarkers, RNA, DNA, pathogen, antigen or portion thereof, antibody, virus, metabolite generated as a reaction to disease or infection, or viral protein.
(32) As used herein, the terms “sample” and “target sample” all refer to any substance that may be subject to the methods and systems provided herein. Particularly, these terms refer to any matter (animate or inanimate) where an analyte may be present and capable of being detected, quantified, monitored or a combination thereof. Suitable examples of targets include, but are not limited to, any animate or inanimate surface, soil, food, ambient air, or soil. Targets also include air, surfaces, fluids and mixtures thereof in or from laboratories, healthcare facilities, human skin, hair or bodily fluids (e.g., whole blood, blood serum, saliva, vaginal fluids, semen, mucus, urine, or similar internal fluid), animal skin, hair or bodily fluid (e.g., whole blood, blood serum, saliva, vaginal fluids, semen, mucus, urine, or similar internal fluid), industrial processes, lakes, rivers, and streams. The target also encompasses exhaled breath.
(33) As used herein, the term “buffer” refers to a carrier that is mixed with the target sample that

includes at least one analyte.
(34) As used herein, the term “test sample composition” refers to the combination of at least one buffer and target sample.
(35) As used herein, the term “healthcare test sample composition” refers to the combination of at least one buffer and target sample taken from a healthcare environment.
(36) As used herein, the term “healthcare environment” refers to a location where patient and medical professional interaction occurs. Such healthcare environments include locations remote from a centralized laboratory facility including, but not limited to, healthcare facilities and remote testing or diagnostic centers (i.e., drive-thru in vitro testing or pop-up in vitro testing center).
(37) As used herein, the term “communication” refers to the movement of air, liquid, mist, fog, buffer, test sample composition, or other suitable source capable of carrying an analyte throughout or within the cartridge system. The term “communication” may also refer to the movement of electronic signals between components both internal and external to the cartridge systems described herein.
(38) As used herein, the term “single-use” refers to the cartridge system being utilized in an interferometric system for a single test or assay before disposal (i.e., not re-used or used for a second time).
(39) As used herein, the term “multiple-use” refers to the cartridge system being utilized for more than one test sample composition (e.g., assay) before disposal.
(40) As used herein, the term “multiplex” refers to the cartridge system being utilized to detect multiple analytes from one target sample composition.
(41) As used herein, the term “pathogen,” “pathological,” “pathological contaminant” and “pathological organism” refer to any bacterium, virus or other microorganism (fungi, protozoa, etc.) that can cause disease for a member of the plant or animal kingdom.
(42) As used herein, the term “point of care” refers to the applicability of the systems provided herein to be utilized by a medical professional or other trained user in various healthcare environments. The systems provided herein may be used by emergency medical technicians while providing care and transport of patients.
(43) In order to address the need for faster and more reliable handling of analyte detection and quantification, portable systems and methods are described herein. Particularly, methods and systems are provided herein to address the need to monitor, identify, quantify, and even certify samples with results provided in a fast, sensitive, and accurate manner. The systems as provided herein may be mobile (hand-held) or portable for ease of point of care use in various environments.
(44) Optical Interferometry Principles
(45) The systems provided include a detector that operates via ultrasensitive, optical waveguide interferometry. The waveguiding and the interferometry techniques are combined to detect, monitor and even measure small changes that occur in an optical beam along a propagation pathway. These changes can result from changes in the length of the beam's path, a change in the wavelength of the light, a change in the refractive index of the media the beam is traveling through, or any combination of these, as shown in Equation 1.
(46) $= 2 \quad Ln /$ Equation 1
(47) According to Equation 1, φ is the phase change, which is directly proportional to the path length, L, and refractive index, n, and inversely proportional to the wavelength (λ) change. According to the systems and methods provided herein, the change in refractive index is used. Optical waveguides are utilized as efficient sensors for detection of refractive index change by probing near the surface region of the sample with an evanescent field. Particularly, the systems provided herein can detect small changes in an interference pattern.
(48) According to one embodiment, the waveguide and interferometer act independently or in tandem to focus an interferometric diffraction pattern. According to one embodiment, the waveguide, interferometer, and sensor act independently or two parts in tandem, or collectively to

focus an interferometric pattern with or without mirrors or other reflective or focal median. According to one embodiment, the waveguide and interferometer exhibit a coupling angle such that focus is at an optimum angle to allow the system to be compact and suited to be portable and hand-held.

(49) Interferometric System Overview

(50) The interferometric systems as provided herein are mobile (hand-held) and portable for ease of use in various environments. The interferometric systems include a weight and overall dimensions such that user may hold the entire interferometric system comfortably in one hand. According to one embodiment, the entire interferometric system is under three pounds. Thus, the present disclosure provides a lightweight, handheld and easy-to-use interferometric system that can rapidly, precisely, and accurately provide detection and quantification of analytes in a variety of environments.

(51) The systems as provided herein provide a high throughput modular design. The systems as provided herein may provide both qualitative and quantitative results from one or more analytes within a test sample composition. Particularly, the systems as provided herein may simultaneously provide detection and quantification of one or more analytes from a target sample. According to one embodiment, both qualitative and quantitative results are provided in real-time or near real time.

(52) The interferometric systems provided herein generally include a housing for various detection, analysis and display components. The interferometric system housing includes a rugged, stable, shell or case. The interferometric system housing can withstand hazards of use and cleaning or disinfection procedures of the case surface. The interferometric system housing may be manufactured from a polymer via various techniques such as injection molding or 3D printing. The interferometric system housing may be manufactured to include a coloration that provides the interferometric system housing with a particular color or color scheme.

(53) According to one embodiment, the interferometric systems provided herein include components that are sealed, waterproof or water resistant to the outside environment to minimize opportunities for contamination of a target sample. The overall arrangement of components within the interferometric systems minimize harboring of contamination in any hard-to-reach areas allowing for ease of disinfection.

(54) The interferometric systems provided herein include a cartridge system. The cartridge systems provided herein integrate with one or more independent or integrated optical waveguide interferometers. The cartridge systems provide efficient sample composition communication through a microfluidic system mounted on or within the cartridge housing. The cartridge is suitable for one or more analytes to be detected in a single sample in a concurrent, simultaneous, sequential or parallel manner. The cartridge systems provided herein may be utilized to analyze in a multiplex manner. That is, one test sample composition will be tested to determine the presence of multiple analytes at the same time by utilizing a plurality of waveguide channels that interact with the test sample composition.

(55) The cartridge systems provided herein are easily removable and disposable allowing for overall quick and efficient use without the risk of cross-contamination from a previous target sample. The cartridge may be safely disposed of after a single use. Disposal after a single use may reduce or eliminate user exposure to biological hazards. According to one embodiment, the cartridge system includes materials that are biodegradable, or recycled materials, to reduce environmental impact. The cartridge system may be cleaned and re-used or otherwise recycled after a single use.

(56) The cartridge system as provided herein may be suited for multiple or one-time use. The single-use cartridge system may be manufactured in a manner such that a buffer solution is pre-loaded in the microfluidic system. By providing the buffer solution pre-loaded in the single-use cartridge system, gas bubbles are reduced or otherwise eliminated. After a single use, the entire

cartridge system is safely discarded or recycled for later use after cleaning. Put another way, after introduction and detection of a test sample composition, the entire single-use cartridge system is not used again and, instead, discarded.
(57) The cartridge systems as provided herein may be suited for multiple uses. According to such an embodiment, the cartridge system may be used one or more times prior to the cartridge system being safely discarded or recycled. The cartridge system may also be cleaned and re-used or otherwise recycled after multiple uses. According to one embodiment, the cartridge system facilitates cleaning and re-tooling to allow the cartridge system to be replenished and returned to operation.
(58) According to one embodiment, the interferometric systems as provided herein have an analyte detection limit down to about 10 picogram/ml. According to one embodiment, the systems as provided herein have an analyte detection limit down to about 1.0 picogram/ml. According to one embodiment, the systems as provided herein have an analyte detection limit down to about 0.1 picogram/ml. According to one embodiment, the systems as provided herein have an analyte detection limit down to about 0.01 picogram/ml.
(59) According to one embodiment, the interferometric systems as provided herein have an analyte detection limit down to about 3000 plaque forming units per milliliter (pfu/ml). According to one embodiment, the systems as provided herein have an analyte detection limit down to about 2000 pfu/ml. According to one embodiment, the systems as provided herein have an analyte detection limit down to about 1000 pfu/ml. According to one embodiment, the systems as provided herein have an analyte detection limit down to about 500 plaque forming units per milliliter (pfu/ml). According to one embodiment, the systems as provided herein have an analyte detection limit down to about 100 plaque forming units per milliliter (pfu/ml). According to one embodiment, the systems as provided herein have an analyte detection limit down to about 10 plaque forming units per milliliter (pfu/ml). According to one embodiment, the systems as provided herein have an analyte detection limit down to about 1 plaque forming units per milliliter (pfu/ml). According to one embodiment, the systems as provided herein have an analyte detection limit to about 1 plaque forming units per liter (pfu/l).
(60) According to one embodiment, the interferometric systems provided herein provide both qualitative and quantitative results at or under 60 minutes after sample introduction to the system. According to one embodiment, both qualitative and quantitative results are provided at or under 30 minutes. According to one embodiment, both qualitative and quantitative results are provided at or under 10 minutes. According to one embodiment, both qualitative and quantitative results are provided at or under 5 minutes. According to one embodiment, both qualitative and quantitative results are provided at or under 2 minutes. According to one embodiment, both qualitative and quantitative results are provided at or under 1 minute.
(61) The interferometric systems as provided herein may be powered via alternating current or direct current. The direct current may be provided by a battery such as, for example, one or more lithium or alkaline batteries. The alternating or direct current may be provided by alternative energy sources such as wind or solar.
(62) According to one embodiment, the interferometric system is stabilized to address vibrational distortions. The system may be stabilized by various means including mechanical, chemically (fluid float or gel pack), computer-assisted system (electronically), or digitally (e.g., via a camera). In some implementations, the systems provided herein allow for point of use assays that are stable in various conditions, including ambient temperature and humidity as well as extreme heat, cold and humidity.
(63) The interferometric systems as provided herein may be equipped with one or more software packages loaded within. The software may be electronically connected to the various system components as provided herein. The software may also be electronically integrated with a display for viewing by a user. The display may be any variety of display types such as, for example, a

LED-backlit LCD. The system may further include a video display unit, such as a liquid crystal display ("LCD"), an organic light emitting diode ("OLED"), a flat panel display, a solid state display, or a cathode ray tube ("CRT").

(64) According to one embodiment, the interferometric system as provided herein may interface with or otherwise communicate with a transmission component. The transmission component may be in electronic signal communication with both the cartridge system and interferometric system components. The transmission component sends or transmits a signal regarding analyte detection data and quantification data. The transmission of such data may include real-time transmission via any of a number of known communication channels, including packet data networks and in any of a number of forms, including instant message, notifications, emails or texts. Such real-time transmission may be sent to a remote destination via a wireless signal. The wireless signal may travel via access to the Internet via a surrounding Wi-Fi network. The wireless signal may also communicate with a remote destination via Bluetooth or other radio frequency transmission. The remote destination may be a smart phone, pad, computer, cloud device, or server. The server may store any data for further analysis and later retrieval. The server may analyze any incoming data using artificial intelligence learning algorithms or specialized pathological, physical, or quantum mechanical expertise programed into the server and transmit a signal.

(65) According to one embodiment, the transmission component may include a wireless data link to a phone line. Alternatively, a wireless data link to a building Local Area Network may be used. The system may also be linked to Telephone Base Unit (TBU) which is designed to physically connect to a phone jack and to provide 900 MHz wireless communications thereby allowing the system to communicate at any time the phone line is available.

(66) According to one embodiment, the interferometric system may include a location means. Such a location means includes one or more geolocation device that records and transmits information regarding location. The location means may be in communication with a server, either from a GPS sensor included in the system or a GPS software function capable of generating the location of the system in cooperation with a cellular or other communication network in communication with the system. According to a particular embodiment, the location means such as a geolocation device (such as GPS) may be utilized from within its own device or from a mobile phone or similarly collocated device or network to determine the physical location of the cartridge system.

(67) According to one embodiment, the interferometric system contains a geo-location capability that is activated when a sample is analyzed to "geo-stamp" the sample results for archival purposes. According to one embodiment, the interferometric system contains a time and date capability that is activated when a sample is analyzed to time stamp the sample results for archival purposes.

(68) The interferometric systems provided herein may interface with software that can process the signals hitting the detector unit. The cartridge system as provided herein may include a storage means for storing data. The storage means is located on or within the cartridge housing or within the interferometric system housing. The storage means communicates directly with electronic components of the interferometric system. The storage means is readable by the interferometric system. Data may be stored as a visible code or an index number for later retrieval by a centralized database allowing for updates to the data to be delivered after the manufacture of the cartridge system. The storage means may include memory configured to store data provided herein.

(69) The data retained in the storage means may relate to a variety of items useful in the function of the interferometric system. According to a particular embodiment, the data may provide the overall interferometric system or cartridge system status such as whether the cartridge system was previously used or is entirely new or un-used. According to a particular embodiment, the data may provide a cartridge system or interferometric system identification. Such an identification may include any series of letter, numbers, or a combination thereof. Such identification may be readable through a QR code. The identification may be alternatively memorialized on a sticker located on the cartridge housing or interferometric system housing. According to one embodiment, the

cartridge housing contains a bar code or QR code. According to one embodiment, the cartridge system contains a bar code or QR code for calibration or alignment. According to one embodiment, the cartridge system contains a bar code or QR code for identification of the cartridge or test assay to be performed. According to one embodiment, the cartridge system contains a bar code or QR code for identification of the owner and location of where any data generated should be transmitted. A user may scan such a QR code with the interferometric system's external camera prior to use to use of the system such that identification and transmission may occur (e.g., automatically or upon user direction).

(70) According to a particular embodiment, the data retained in the storage means may provide the number of uses remaining for a multiple-use cartridge system. According to a particular embodiment, the data may provide calibration data required by interferometric system to process any raw data into interpretable results. According to a particular embodiment, such data may relate to information about the analyte and any special processing instructions that can be utilized by the cartridge system to customize the procedure for the specific combination of receptive surface(s) and analyte(s). The interferometric system as provided herein may include electronic memory to store data via a code or an index number for later retrieval by a centralized database allowing for updates to the data to be delivered after the manufacture of the cartridge system.

(71) The interferometric system may include a memory component such that operating instructions for the interferometric system may be stored. All data may be stored or archived for later retrieval or downloading onto a workstation, pad, smartphone or other device. According to one embodiment, any data obtained from the system provided herein may be submitted wirelessly to a remote server. The interferometric system may include logic stored in local memory to interpret the raw data and findings directly, or the system may communicate over a network with a remotely located server to transfer the raw data or findings and request interpretation by logic located at the server. The interferometric system may be configured to translate information into electrical signals or data in a predetermined format and to transmit the electrical signals or data over a wireless (e.g., Bluetooth) or wired connection within the system or to a separate mobile device. The interferometric system may perform some or all of any data adjustment necessary, for example adjustments to the sensed information based on analyte type or age, or may simply pass the data on for transmission to a separate device for display or further processing.

(72) The interferometric systems provided herein may include a processor, such as a central processing unit ("CPU"), a graphics processing unit ("GPU"), or both. Moreover, the system can include a main memory and a static memory that can communicate with each other via a bus. Additionally, the system may include one or more input devices, such as a keyboard, touchpad, tactile button pad, scanner, digital camera or audio input device, and a cursor control device such as a mouse. The system can include a signal generation device, such as a speaker or remote control, and a network interface device.

(73) According to one embodiment, the interferometric system may include color indication means to provide a visible color change to identify a particular analyte. According to one embodiment, the system may include a reference component that provides secondary confirmation that the system is working properly. Such secondary confirmation may include a visual confirmation or analyte reference that is detected and measured by the detector.

(74) The interferometric system as provided herein may also include a transmitting component. The transmitting component may be in electronic signal communication with the detector component. The transmitting component sends or transmits a signal regarding analyte detection and quantification data. The transmission of such data may include real-time transmission via any of a number of known communication channels, including packet data networks and in any of a number of forms, including text messages, email, and so forth. Such real-time transmission may be sent to a remote destination via a wireless signal. The wireless signal may travel via access to the Internet via a surrounding Wi-Fi network. The wireless signal may also communicate with a remote

destination via Bluetooth or other radio frequency transmission. The remote destination may be a smart phone, pad, computer, cloud device, or server. The server may store any data for further analysis and later retrieval. The server may analyze any incoming data using artificial intelligence learning algorithms or specialized pathological, physical, or quantum mechanical expertise programed into the server and transmit a signal.

(75) According to one embodiment, the interferometric system includes a wireless data link to a phone line. Alternatively, a wireless data link to a building Local Area Network may be used. The system may also be linked to Telephone Base Unit (TBU) which is designed to physically connect to a phone jack and to provide 900 MHz wireless communications thereby allowing the system to communicate at any time the phone line is available.

(76) According to one embodiment, the system may also include geolocation information in its communications with the server, either from a GPS sensor included in the system or a GPS software function capable of generating the location of the system in cooperation with a cellular or other communication network in communication with the system. According to a particular embodiment, the system may include a geolocation device (such as GPS or RFID) either from within its own device or from a mobile phone or similarly collocated device or network to determine the physical location of the system.

(77) According to one embodiment, the interferometric system includes an external camera. The external camera may be at least partially located within the interferometric system housing but include a lens exposed to the exterior of the housing such that the external camera may take photos and video of a target sample prior to collection (e.g., soil, plant, etc.). The external camera may capture video or images that aid in the identification of an analyte and confirmation of the resulting data. The external camera may also capture video images that aid in selecting a proper remedial measure. The external camera may capture video or images that aid in the identification of a target sample or source thereof.

(78) The external camera may capture video or images in connection with scanning and identifying a QR code (such as a QR code on an external surface of a cartridge housing). When located on an external surface of the cartridge housing, the QR code may also aid in identifying ownership of generated data and transmission of such data to a correct owner.

(79) According to one embodiment, the cartridge system contains a geo-location capability that is activated when a sample is analyzed to “geo-stamp” the sample results for archival purposes. According to one embodiment, the cartridge system contains a time and date capability that is activated when a sample is analyzed to time stamp the sample results for archival purposes. According to one embodiment, the cartridge system includes materials that are biodegradable, or recycled materials, to reduce environmental impact. Any used cartridge system provided herein may be disposed of in any acceptable manner such as via a standard biohazard container. According to one embodiment, the cartridge system facilitates cleaning and re-tooling to allow the cartridge system to be replenished and returned to operation.

(80) According to one embodiment, the cartridge system is stabilized to address vibrational distortions. The system may be stabilized by various stabilization means including mechanical (alignment means as provided herein), chemically (fluid float or gel pack), computer-assisted system (electronically), or digitally (e.g., via a camera or digital processing).

(81) Microfluidic System Overview—Single-Use Cartridge System

(82) The single-use cartridge system provided herein includes a microfluidic system for communicating or otherwise providing a means for test sample and buffer to mix thereby resulting in a test sample composition. The microfluidic system causes the test sample composition to move through the detection region to allow for detection and analysis of one or more analytes. The microfluidic system includes an injection port for introduction of a test sample. The injection port may optionally include a check valve. The microfluidic system further includes a first microchannel section having a first end attached in communication with the injection port check

valve and a second end in communication with a mixing bladder. According to one embodiment, the first microchannel section contains a filter to remove materials not capable of detection and quantification. The mixing bladder is sized, shaped and otherwise configured to store buffer. The mixing bladder is sized, shaped and otherwise configured to aid in mixing buffer and test sample to form the test sample composition. The mixing bladder may be bypassed such that the test sample composition may be automatically discharged or allowed to proceed through the microfluidic system. The mixing bladder may include a temperature control means in the form of a metal coil wrapped around the mixing bladder such that the temperature control means is heated upon introduction of an electric current.

(83) The microfluidic system further includes second microchannel section having a first end attached in communication with the mixing bladder and a second end attached in communication with a flow cell having at least one detection microchannel. By including multiple two or more detection microchannels, the cartridge system is particularly suited for high throughput and improved testing efficiency by being able to detect and quantify analyte in more than one test sample composition.

(84) The microfluidic system further includes at least one pump. Suitable pumps include micropumps such as, but are not limited to, diaphragm, piezoelectric, peristaltic, valveless, capillary, chemically-powered, or light-powered micropumps. According to an alternative embodiment, the microfluidic system further includes at least one pump that is a, positive-displacement pump, impulse pump, velocity pump, gravity pump, steam pump, or valve-less pump of any appropriate size. According to a single-use embodiment of the cartridge system, the cartridge system contains at least one pump located within the cartridge housing. According to one embodiment of a single-use cartridge system, the pump overlays or otherwise engages or touches the first microchannel section, second microchannel section and mixing bladder.

(85) The microfluidic system of the single-use cartridge system as provided herein may be manufactured and packaged under negative pressure or vacuum sealed. In this manner, the negative pressure allows for a test sample to be pulled in and become self-loading upon introduction of the test sample. The negative pressure further allows for a test sample to be pulled in in the microfluidic system to reduce, avoid or eliminate bubble formation upon introduction of the test sample. According to an alternative embodiment, the microfluidic system is manufactured and packaged under a positive pressure. According to either embodiment, the microfluidic system of a single-use cartridge system may be pre-loaded with a buffer solution at the time of manufacture. The buffer may be custom designed or designated for a particular analyte detection. Buffer solution that is used (i.e., buffer waste) and resulting test sample composition waste may be contained permanently in the single-use cartridge system.

(86) According to one embodiment, the pump can be powered by a battery or electricity transferred from the testing device. Alternatively, the energy to power the pump can be mechanically transferred by direct force, electromagnetic induction, magnetic attraction, audio waves, or piezo electric transfer. According to one embodiment, the cartridge system includes at least one pulse dampening component such as a regulator or accumulator or bladder.

(87) Microfluidic System Overview—Multiple-Use Cartridge System

(88) The multiple-use cartridge system provided herein includes a microfluidic system for communicating or otherwise providing a means for a test sample composition to move through the cartridge system and allow for detection and analysis of one or more analytes. According to a particular embodiment, the test sample and test sample composition are air or liquid. An ingress port is located on a front surface of the multiple-use cartridge system. The ingress port is in communication with a first microchannel section having a first end attached in communication with an ingress port check valve and a second end in communication with second microchannel section. A filter may be located anywhere within the first microchannel section.

(89) The second microchannel section includes a first end in communication the first microchannel

section and a second end in communication with a flow cell having at least one detection microchannel. The cartridge system includes a detection region that accommodates or is otherwise adapted to receive the chip and flow cell wafer.

(90) The detection microchannel is in communication with a first end of a third microchannel section. The third microchannel section includes a flow electrode to approximate flow rate and is correlated with measured impedance. The third microchannel section includes a second end in communication with the first end of a fourth microchannel. The fourth microchannel includes a second end in communication with a check valve which, in turn, is in communication with an egress port. The chip utilized in the multiple-use embodiment may be removable from the cartridge system.

(91) The microfluidic system further includes at least one pump. Suitable pumps include micropumps that include, but are not limited to, diaphragm, piezoelectric, peristaltic, valveless, capillary, chemically-powered, or light-powered micropumps. According to an alternative embodiment, the microfluidic system further includes at least one pump that is a positive-displacement pump, impulse pump, velocity pump, gravity pump, steam pump, or valve-less pump of any appropriate size. According to one multiple-use embodiment of the cartridge system, the cartridge system contains at least one pump located outside (external to) the cartridge housing but in communication with the microfluidic system. The external pump may be utilized to move test sample composition through the microfluidic system to aid in removal of air or bubble that may be present in a liquid test sample composition prior to use. According to one embodiment, the cartridge system contains at least one pump dampening device.

(92) All of the cartridge systems provided herein may utilize the pump to manipulate the communication of test sample composition throughout the microfluidic system. According to one embodiment, the pump causes or otherwise aids movement of test sample composition through the microchannels as well as the mixing bladder, when present.

(93) Handheld Interferometric System—Exemplary Embodiment

(94) FIG. **1** illustrates a perspective view of one embodiment of a portable interferometric system **100** as provided herein. The portable interferometric system **100** may include a display unit **102**. The portable interferometric system **100** may include a housing **104** adapted to fit within a user's hand.

(95) FIG. **2**A illustrates a front view of one embodiment of a portable interferometric system **100** that utilizes the cartridge systems provided herein. The housing **104** includes an external front surface **106** defining an opening **108** adapted to receive the cartridge system provided herein. The opening **108** aids in the alignment and proper position of the cartridge system as provided herein within the handheld interferometric system **100**. The opening **108** may optionally include a flap **110** that shields or covers the opening **108** when the cartridge is not inserted. The flap **110** may be hinged on any side so as to aid in the movement of the flap **110** from a first, closed position to a second, open position upon insertion of the cartridge system.

(96) FIG. **2**B illustrates a rear view of one embodiment of a portable interferometric system **100** as provided herein. The housing **104** is adapted to include USB Type C **112**, USB Type A **114**, data or phone line inlet **116**, power cord inlet **118**, power switch **120**, and external camera or other light sensitive device **122**.

(97) Chip Overview

(98) As previously noted, the cartridge systems provided herein further includes a detection region. This detection region accommodates or is otherwise adapted to receive an interferometric chip and flow cell wafer. The flow cell wafer includes at least one detection microchannel. The flow cell wafer is located directly above the chip. The detection microchannel may be etched onto a flow cell wafer having a substantially transparent or clear panel or window. The detection microchannel aligns with each waveguide channel in the chip.

(99) In use, a light signal may be emitted from a light unit located in the interferometric system.

The light enters flow through entry gradients in the chip and through one or more waveguide channels. According to a particular embodiment, there may be two or more waveguide channels to determine the presence of a separate analyte that each of the individual wavequide channels alone would not have been able to identify alone. The evanescent field is created when the light illuminates the waveguide channel. The light signal is then directed by exit gradients to a detector unit such as a camera unit. The detector unit is configured to receive the light signal and detect an analyte present in a test sample composition. The chip may further include a reference waveguide channel.
(100) A sensing layer is adhered to a top side of one or more waveguide channels. According to a particular embodiment, the sensing layer may include one or more antigens or antibodies that are immobilized on the waveguide channel surface to sense the antigen-specific antibody or antigen, respectively. According to another embodiment, the sensing layer may include envelope, membrane, nucleocapsid N-proteins or different domains of one of the proteins in a natural or artificial virus used to deliver interfering RNA (RNAi) as a treatment.
(101) According to a particular embodiment, the sensing layer may include a molecularly imprinted polymer. The molecularly imprinted polymer leaves cavities in the polymer matrix with an affinity for a particular analyte such as an antibiotic.
(102) According to a particular embodiment, the sensing layer may include a DNA microarray of DNA probes. Each probe may be specific for a pathogen (i.e., bacterial species) and when the probe hybridizes with a sample, the sample/probe complex fluoresces in UV light or may be detected via interferometric analysis.
(103) According to one embodiment, the sensing layer may utilize immunoassays on top of the waveguide channels for detection of one or more analytes. According to one embodiment, the system may include, or function based on, an enzyme-linked immunosorbent assay (ELISA) or other ligand binding assays that detect analytes in target samples. According to one embodiment, the sensing layer may utilize one or more polypeptides, nucleic acids, antibodies, carbohydrates, lipids, receptors, or ligands of receptors, fragments thereof, and combinations thereof. According to one such embodiment, the sensing layer is configured to include one or more antibodies as well as one or more immunoglobulins to aid in the indication of the stage of analyte infection. Suitable immunoglobulins include IgG, IgM, IgA, IgE and IgD.
(104) Flow Cell Overview
(105) Each of the cartridge systems described herein include a flow cell having at least one detection microchannel adapted to communicate with one or more test sample compositions flowing through a waveguide channel in a chip beneath the flow cell. According to one embodiment, the cartridge systems may include at least two, at least three, or at least four detection microchannels with each detection microchannel adapted to communicate one or more test sample composition allowing detection of the same or different analytes.
(106) Each detection microchannel is located on or within a flow cell manufactured from a wafer. The at least one detection microchannel may be etched, molded or otherwise engraved into one side of the flow cell wafer. Thus, the at least one detection microchannel may be shaped as a concave path as a result of the etching or molding within the flow cell wafer.
(107) The flow cell wafer is oriented above the chip during use such that the detection microchannel may be orientated or otherwise laid out in variety of flow patterns above the waveguide channels. The detection microchannel may be laid out, for example, in a simple half loop flow pattern, serial flow pattern, or in a serpentine flow pattern. The serpentine flow pattern is particularly suited for embodiments where there are multiple waveguide channels that are arranged in a parallel arrangement. By utilizing the serpentine flow pattern, the test composition flows consistently over the waveguide channels without varying flow dynamics.
(108) Chip, Flow Cell and Optical Assembly—Exemplary Embodiment
(109) FIG. **3**A illustrates a cross-sectional view of an optical detection region **200** of a cartridge

system. A chip (or substrate) **202** includes a waveguide channel **204** attached to a surface **205** (such as the illustrated top surface) of the chip **202**. An evanescent field **206** is located above the waveguide channel **204**. A sensing layer **208** is adhered to a top side of the waveguide channel **204**. As illustrated, antibodies **210** are shown that may bind or otherwise immobilized to the sensing layer **208**, however, the sensing layer **208** may be adapted to bind any variety of analytes. As such, adjusting or otherwise modifying the sensing layer **208** allows for the cartridge system to be utilized for multiple different types of analytes without having to modify the cartridge system or and surrounding interferometric system components. In general use, a light signal (e.g., laser beam) illuminates the waveguide channel **204** creating the evanescent field **206** that encompasses the sensing layer **208**. Binding of an analyte impacts the effective index of refraction of the waveguide channel **204**.

(110) A bottom view of an exemplary flow cell **300** is illustrated in FIG. **3**B. At least one detection microchannel **302** is located on or within a flow cell **300** manufactured from a transparent wafer. The at least one detection microchannel **302** may be etched, molded or otherwise engraved into one side of the flow cell wafer **304**. Thus, the at least one detection microchannel **302** may be shaped as a concave path as a result of the etching or molding within the flow cell wafer **304**. The flow cell wafer **304** may be manufactured a material such as opaque plastic, or other suitable material. The flow cell wafer **304** may optionally be coated with an anti-reflection composition.

(111) The movement of a light signal **308** (series of arrows) through a chip **310** is illustrated in FIG. **3**C. The light signal **308** moves from a light unit **312**, such as a laser unit, through a plurality of entry gradients **314** and through one or more waveguide channels **316**. Each channel includes a pair of waveguides (**321**, **323**). One of the pair of waveguides **321** is coated with a sensing layer **208** (as indicated by shading in FIG. **3**C). The other one of the pair of waveguides **323** is not coated with the sensing layer **208** (serving as a reference). The combination of the light from each in the pair of waveguides (**312**, **323**) creates an interference pattern which is imaged onto detector unit **320**.

(112) According to a particular embodiment, the two or more waveguide channels **316** are utilized that are able to determine the presence of an analyte that each of the individual waveguide channels **316** alone would not have been able to identify alone. The light signal **308** is then directed by exit gradients **318** to a detector unit **320** such as a camera unit. The detector unit **320** is configured to receive the light signal **308** and detect any analyte present in a target sample composition flowing through the detection microchannel **302** (see FIG. **3**B).

(113) The chip **310** includes a combination of substrate **202** (see FIG. **3**A), waveguide channel (see FIG. **3**A part **204** and FIG. **3**C part **316**) and sensitive layer **208** (see FIG. **3**A). The flow cell **300** is oriented above the top surface **205** of the chip **310** during use such that the detection microchannel **302** may be orientated or otherwise laid out in variety of flow patterns above the waveguide channels **316**. The detection microchannel **302** may be laid out, for example, in a simple half loop flow pattern, serial flow pattern, or in a serpentine flow pattern as illustrated in FIG. **3**B. The serpentine flow pattern is particularly suited for embodiments where there are multiple waveguide channels **316** that are arranged in a parallel arrangement (see FIG. **3**C). By utilizing the serpentine flow pattern, the test composition flows consistently over the waveguide channels **316** without varying flow dynamics.

(114) The light signal passes through each waveguide channel as illustrated in FIG. **3**C, and may combine thereby forming diffraction patterns on the detector unit. The interaction of the analyte **210** (see FIG. **3**A) and the sensing layer **208** changes the index of refraction of light in the waveguide channel per Equation 1. The diffraction pattern is moved which is detected by the detector unit. The detector unit as provided herein may be in electronic communication with video processing software. Any diffraction pattern movement may be reported in radians of shift. The processing software may record this shift as a positive result. The rate of change in radians that happens as testing is conducted may be proportional to the concentration of the analyte.

(115) FIG. **4** illustrates a side view of an exemplary embodiment of an optical assembly unit **400**

that can be found in the handheld interferometric systems described herein (such as in FIGS. **1-2**). The optical assembly unit **400** includes a light unit **402** aligned in a light unit housing **404**. The optical assembly unit **400** includes a detector unit **406**, such as a camera unit, aligned in a camera unit housing **408**.

(116) FIG. **5**A illustrates a cross-sectional view of the optical assembly unit **400** of FIG. **4**. The light unit **402** is situated at an angle relative to the shutter flap element **420**. The shutter flap element **420** is adapted to slide open and shut under tension from a shutter spring **422**. The shutter flap element **420** is illustrated in a first, closed position with no cartridge system inserted. The shutter flap element **420** includes and upper control arm **423** that is located within a rail portion **425**.

(117) A complimentary communication means **424** extends downward so as to make electronic contact with electronic communications means located on the cartridge housing (see FIGS. **6**, **8**A and **9**A). The complimentary communication means **424** may be metal contacts such that, upon insertion, the metal contacts on the exterior surface of the cartridge housing touch and establish electronic communication between the cartridge system and the remaining components of the interferometric system (e.g., light unit, camera unit, etc.). The complimentary communication means **424**, as illustrated, include one or more substantially pointed or “V” shaped so as to push down into or otherwise contact the cartridge housing metal contacts. The number of complimentary communication means **424** may match and align with the number of metal contacts on the exterior surface of the cartridge housing.

(118) At least one downward cantilever bias spring **426** may be located within the optical assembly unit **400** such that, upon insertion of the cartridge through the interferometric system housing opening, the downward cantilever bias spring **426** pushes against a top side of the cartridge housing thereby forcing the cartridge housing against an opposite side or bottom portion or surface **428** of the cartridge recess **430** resulting in proper alignment along a vertical plane (see FIGS. **5**A, **5**B, **5**C and **6**).

(119) The light unit **402** is optionally adjustable along various planes for optimal light signal **432** emission. As illustrated, the signal **432** is shown to be emitted and focused by at least one lens **433**. A camera unit **406** is situated at an angle relative to the shutter flap element **420** so as to receive the light signal **432** upon exit from the cartridge (see FIG. **6**).

(120) A first roll adjustment screw **434** and second roll adjustment screw **436** are located on opposing sides of the light unit **402** for adjusting roll of the light unit **402**. A first upward adjustment screw **438** and second upward adjustment screw **440** are located in a parallel manner on each side the light unit **402** for adjusting the light unit **402** towards the cartridge system (i.e., substantially upward). An angle of incidence screw **442** is located against the light unit **402** to allow for adjustments to the angle of incidence for proper coupling angle. A translation screw **444** is located direct communication with the light unit **402** to adjust translation in the X axis. A spring element **446** maintains the position of the light unit **402** against the light unite **402** by assisting the adjustment screws (**432**, **436**), incidence screw **442** and translation screw **444**.

(121) With specific regard to FIGS. **5**A, **5**B, and **5**C, the bottom portion **428** of the cartridge recess **430** further includes alignment means that includes at least one rail portion **425** for engaging both male key portions on the cartridge housing (see **824**, **826** of FIG. **8**A; see **920**, **922** of FIG. **9**A). The bottom portion or surface **428** of the cartridge recess **430** includes a first raised surface **421**A and second raised surface **421**B. A shutter upper control arm **423** is located within the rail portion **425**. The rail portion **425** includes a first rail wing **427** and second rail wing **429** adapted to receive and engage the male key portions (see **824**, **826** of FIG. **8**A; see **920**, **922** of FIG. **9**A). By including such alignment means, the cartridge systems provided here may only engage in a certain manner thereby preventing incorrect insertion and provided proper optical and microfluidic alignment.

(122) FIG. **6** illustrates a cross-sectional view of the optical assembly **400** of FIG. **5**A with one

embodiment of a cartridge system **800** inserted in the optical assembly **400**. As illustrated, the shutter flap element **420** is pushed backwards upon insertion of the cartridge system **800**. While not shown, the shutter spring **422** is compressed backwards. The shutter flap element **420** moves along a track system **450** having a stationary male rail **452** on which a female rail portion **454** slides from a first, closed position with no cartridge system **800** inserted to a second, open position as illustrated in FIG. **6** upon cartridge system **800** insertion.
(123) FIG. **6** further illustrates positioning of the cartridge system **800** in the optical assembly **400**. The cartridge system **800** includes an interferometric chip **832** positioned below the flow cell wafer **888**. The cartridge system **800** includes storage means **807** as provided herein positioned within the cartridge housing **802**. While the cartridge system **800** is illustrated as a single-use system, the alignment and positioning of the single-use cartridge assembly may also apply to the multiple-use cartridge systems provided herein (e.g., see FIGS. **9**A-**9**F).
(124) FIG. **7** illustrates a top view of the optical assembly unit **400** of FIG. **5**A with one embodiment of a cartridge system **800** inserted in the optical assembly unit **400**. The cartridge system **800**, as illustrated, is a single-use system, however, a multiple-use system may be inserted in the same manner within the interferometric system. The cartridge system **800** includes a cartridge housing **802** having a top surface **805**. The optical assembly unit **400**, as illustrated, includes a plurality of cantilever bias springs **426**. The optical assembly unit **400** further includes at least one side bias spring **460** such that, upon insertion of the cartridge system **800**, the side bias spring **460** pushes against one horizontal side **860** of the cartridge housing thereby forcing the cartridge housing **802** into proper alignment along a horizontal plane.
(125) Cartridge System Overview
(126) The cartridge systems provided herein includes a cartridge housing. The cartridge housing may be manufactured from any polymer suitable for single or multiple-use. The cartridge may be manufactured according to a variety of additive processing techniques such as 3-D printing. The cartridge may be manufactured via traditional techniques such as injection molding. The polymer may include a coefficient of expansion such that the housing does not expand or contract in a manner that would disrupt alignment of any microfluidic or detection components described herein when the cartridge is exposed to heat or cold environmental conditions.
(127) The cartridge housing may include a light prevention means to aid in reducing, preventing or eliminating ambient, outside light from interfering the detection of one or more analytes. The light prevention means may include colored cartridge housing (e.g., black colored) that is color dyed or coated during manufacture. According to one embodiment, a dye may be introduced to the polymer to provide a specific color to a region of or the entire cartridge housing. Suitable colors include any color that aids in reducing, preventing or eliminating ambient, outside light from interfering the detection of one or more analytes.
(128) The cartridge systems provided herein further includes a detection region. This detection region accommodates or is otherwise adapted to receive an interferometric chip and flow cell wafer. The flow cell wafer includes at least one detection microchannel. The flow cell wafer is located directly above the chip. The detection microchannel may be etched onto a flow cell wafer having a substantially transparent or clear panel or window. The flow cell wafer, the chip or both the flow cell and chip may be coated with a substance that reduces or eliminates fogging or condensation. According to one embodiment, the chip may be heated to reduce or elimination fogging or condensation.
(129) The cartridge systems provided herein are configured or otherwise adapted or designed to easily insert and instantly align within an interferometric system such as, for example, a hand-held interferometric system. By being configured to allow for instant alignment, no further adjustment is required by a user to align any microfluidic components and any internal detection-related components such as the laser, chip with waveguides and exposed channels in a detection region of the cartridge, optical detector and any other focus-related components in the interferometric

system.
(130) The cartridge housing includes dimensions that are complimentary in size and shape to the size and shape to an internal surface defining a recess within an interferometric system. As provided and illustrated in the non-limiting examples herein, the cartridge housing may be generally rectangular in overall shape.
(131) According to one embodiment, the cartridge system may be inserted and removed automatically. According to one embodiment, the cartridge housing contains a bar code or QR code. According to one embodiment, the cartridge system contains a bar code or QR code for calibration or alignment.
(132) To aid in alignment, the cartridge housing includes an alignment means on an external surface of the cartridge housing. The alignment means many take a variety of forms that assure instant alignment of any microfluidic components and any internal detection-related components upon insertion of the cartridge within the interferometric system. The alignment means also aids in the prevention of incorrect orientation assertion within the interferometric system and allows for insertion only after proper alignment is attained. The alignment means further allows for the cartridge system to be stabilized to address vibrational distortions.
(133) The alignment means may include at least one male key portion for engaging and securing within a corresponding female rail located in the interferometric system. The male key portion may be disposed on the bottom surface of the cartridge housing, however, the male key portion may be located on any exterior surface of the cartridge housing. Other suitable alignment means include one or more microswitches or sensing devices that guide the cartridge housing to assure proper alignment.
(134) According to a particular embodiment, the cartridge housing includes a top portion and a bottom portion based on the orientation of insertion in an interferometric system. The top portion may include a top surface defining at least one through hole on at least one external surface of either the top portion or bottom portion. The at least one through hole is adapted to receive a removable fastening means for securing the top portion and bottom portion together. Suitable fastening means include screws or other suitable fastener that may be removed. By allowing the top portion and bottom portion of the cartridge housing to be separated and re-attached, a user may open the cartridge housing to allow for cleaning as well as replacement of the chip.
(135) The cartridge system as provided herein may include a temperature control means to control temperature as well as humidity. The cartridge system as provided herein may include a temperature control means to control test sample composition temperature. By controlling temperature and humidity around the cartridge system, the interferometric system can provide more repeatable, precise results. According to one embodiment, the cartridge system contains heating capability to facilitate consistent measurement and operation in cold temperatures. By controlling temperature and humidity around the cartridge system, fogging or condensation that causes interference in the detection region of the cartridge system is reduced or otherwise eliminated. The temperature control means may be located on or within the cartridge housing. According to a single-use cartridge system embodiment, the temperature control means is located on or around the mixing bladder of the microfluidic fluid system described herein. The temperature control means may be located on an exterior surface of the cartridge housing. One suitable temperature control means includes a metal coil that is heated upon introduction of an electric current. Another suitable temperature control means includes one or more warming bands or Peltier devices that can provide heating or cooling.
(136) Each of the cartridge systems described herein include a flow cell having at least one detection microchannel adapted to communicate with one or more test sample compositions flowing through a waveguide channel in a chip beneath the flow cell. According to one embodiment, the cartridge systems may include at least two detection microchannels with each detection microchannel adapted to communicate one or more test sample composition allowing

detection of the same or different analytes. According to one embodiment, cartridge system includes a flow cell having at least three detection microchannels with each detection microchannel adapted to communicate one or more test sample composition allowing detection of the same or different analytes. According to one embodiment, cartridge system includes a flow cell having at least four detection microchannels with each detection microchannel adapted to communicate one or more test sample composition allowing detection of the same or different analytes.
(137) Cartridge System—Exemplary Embodiments
(138) An exemplary embodiment of a single-use cartridge system **800** is illustrated in FIGS. **8**A-F. A top view of a cartridge system **800** is provided in FIG. **8**A. The cartridge system **800** includes a cartridge housing **802** as described herein. The housing **802** includes a top portion **804** (see FIG. **8**C) having a top surface **805**. The top surface **805** includes four heat stake posts **808** for joining the top portion **804** of the cartridge housing **802** to a bottom portion **810** (See FIG. **8**C) of the cartridge housing **802**. By utilizing heat stake posts **808**, the top portion **804** may be permanently joined to a bottom portion **810** of the cartridge housing **802**. The top surface **805** includes an injection port **812** for introduction of a test sample.
(139) The cartridge housing **802** further includes an electronic communication means **816** located on a second external surface **818** that is on a different horizontal plane from the top surface **805**. The electronic communication means **816** as illustrated includes a plurality of metal contacts.
(140) The cartridge system further includes a vent port **820**. The vent port **820** allows for any air in the microfluidic system **870** (see FIG. **8**F), such as in the form of bubbles, to exit. The vent port **820** may include a vent cover **821** over the vent port **820**. The vent cover **821** may be fabricated from a material that repels liquid while allowing air or vapor to pass through such as, for example, expanded polytetrafluoroethylene (commercially available as Goretex®. The vent cover **821** allows for air purging from the cartridge system **800** but will not allow fluid to pass through such as when a vacuum is applied to prime the microfluidic system **870**. In this way, bubble formation in a liquid test sample composition is removed or otherwise avoided. The top surface **805** also includes two port seals **822**. The port seals **822** may be made from rubber and provides sealing of the microfluidic system **870** within the cartridge system **800**.
(141) FIG. **8**B illustrates a view of the bottom surface **823** of one embodiment of a single-use cartridge system **800**. The bottom surface **823** includes a first male key portion **824** and a second male key portion **826**. The male keying portions (**824**, **826**) engage with a corresponding rail portion (**425**—See FIGS. **5**A, **5**B and **5**C) located in the cartridge recess **430** of the optical assembly **400**. The bottom surface **823** further defines a first detent **828** and a second detent **830**. The detents (**828**, **830**) engage with or otherwise receive a corresponding first raised surface and a second raised surface (**421**A, **421**B) inside the cartridge recess **430** of the optical assembly **400** (see FIGS. **5**A, **5**B and **5**C). When engaged with the first detent **828** and second detent **830**, the first raised surface and second raised surface (**421**A, **421**B) aid in securing the cartridge system **800** within the cartridge recess **430**.
(142) The chip **832** is substantially transparent and allows the light signal to enter, interact with one or more waveguides channels (See FIG. **3**C) and allow for binding of analyte flowing within the at least one detection microchannel **834** within the flow cell wafer **888**.
(143) The bottom surface **823** further defines a light inlet slot **836**. The light inlet slot **836** allows for a light signal to enter the cartridge system **800**. Particularly, the light inlet slot **836** allows for a light signal to enter the chip **832** and for the light signal to move through any waveguide channels (not shown; see e.g., part **316** of FIG. **3**C) in the chip **832** while interacting with analytes in the at least one detection microchannel **834** before the light signal is deflected by one or more gratings (not shown) down to the detector unit **406** (see e.g., FIG. **3**C and **6**).
(144) FIG. **8**C illustrates a view of the back surface **840** of the cartridge housing **802** of a single-use cartridge system **800**. The cartridge housing **802** includes a top portion **804** and a bottom portion **810**. The male keying portions (**824**, **826**) are shown extending from the bottom portion **810** of the

cartridge housing **802**.

(145) FIG. **8**D illustrates a view of the front surface **850** of the cartridge housing **802** of a single-use cartridge system **800**. The male keying portions (**824**, **826**) are shown extending from the bottom portion **810** of the cartridge housing **802**.

(146) FIG. **8**E illustrates a view of one side surface **860** of the cartridge housing **802** of a single-use cartridge system **800**, the opposing side being a mirror image.

(147) FIG. **8**F illustrates a cross-section view downward of a single-use cartridge system **800** along the horizontal line of FIG. **8**E. The cartridge system **800** includes a detection region **831** that accommodates or is otherwise adapted to receive a chip **832** and flow cell wafer **888**. The single-use cartridge system **800** includes a microfluidic system **870** for communicating or otherwise providing a means for a test sample composition to move through the cartridge system **800** and allow for detection and analysis of one or more analytes. The microfluidic system **870** includes an injection port **812** for introduction of a test sample. The injection port may **812** optionally include a check valve **872**. The microfluidic system **870** further includes a first microchannel section **874** having a first end **876** attached in communication with the injection port check valve **872** and a second end **878** in communication with a mixing bladder **880**. A filter **877** may be located anywhere within the first microchannel section **874**. The microfluidic system **870** also includes a vent port **820** within the first microchannel section **874** between the first end **876** and second end **878**. The mixing bladder **880** includes a temperature control means **881** in the form of a metal coil wrapped around the mixing bladder **880** such that the temperature control means **881** is heated upon introduction of an electric current.

(148) The microfluidic system **870** further includes second microchannel section **882** having a first end **884** attached in communication with the mixing bladder **880** and a second end **886** attached in communication with a flow cell wafer **888** having at least one detection microchannel **834**.

(149) The microfluidic system **870** further includes third microchannel section **890** having a first end **892** attached in communication with at least one detection microchannel **834** and a second end **894** in communication back to the mixing bladder **880** so as to form a closed loop.

(150) The microfluidic system **870** further includes at least one micropump **898**. The micropump **898**, as illustrated, is a piezoelectric pump that overlays or otherwise engages or touches one or more of the first microchannel section **874**, second microchannel section **882**, third microchannel section **890** and mixing bladder **880**. The micropump **898** manipulates the communication of test sample composition throughout the microfluidic system **870**.

(151) The single-use cartridge system **800** may further include a transmission component **897** as provided herein. The single-use cartridge system **800** may further include a location means **899** as provided herein.

(152) An exemplary embodiment of a multiple-use cartridge system **900** is illustrated in FIGS. **9**A-F.

(153) A top view of an embodiment of a multi-use cartridge system **900** is provided in FIG. **9**A. The cartridge system **900** includes a cartridge housing **902** as described herein. The housing **902** includes a top portion **904** (see FIG. **9**C) having a top surface **905**. As illustrated, the top surface **905** includes four top through holes **908**A. The top through holes **908**A are adapted (e.g., threaded) to receive a removable fastening means (not shown) for securing the top portion **904** to a bottom portion **910** (see FIG. **9**C). The top surface also includes two sealing holes **908**B that allow for sealing of the chip **936** to the cartridge housing **902**.

(154) The cartridge housing **902** further includes an electronic communication means **916** located on a second external surface **918** that is on a different horizontal plane from the top surface **905**. The electronic communication means **916** as illustrated includes a plurality of metal contacts. The top surface **905** also includes two port seals **919** and two seal plugs (**924**, **926**).

(155) FIG. **9**B illustrates a view of the bottom surface **923** of a multiple-use cartridge system **900**. The bottom surface **923** includes a first male key portion **920** and a second male key portion **922**.

The male keying portions (**920**, **922**) engage with a corresponding rail portion (**425**—See FIGS. **5**A, **5**B and **5**C) located in the interferometric system. The bottom surface **923** further defines a first detent **928** and a second detent **930**. The detents (**928**, **930**) engage with or otherwise receive a corresponding first raised surface and a second raised surface (**421**A, **421**B) inside the cartridge recess **430** of the optical assembly **400** (see FIGS. **5**A, **5**B and **5**C). When engaged with the first detent **928** and second detent **930**, the first raised surface and second raised surface (**421**A, **421**B) aid in securing the cartridge system **900** within the cartridge recess **430**.

(156) The bottom surface further includes bottom through holes **908**C that align and correspond to the four top through holes **908**A. The bottom through holes **908**C may be adapted (e.g., threaded) to receive a removable fastening means (not shown) for securing the top portion **904** to a bottom portion **910** (see FIG. **9**C).

(157) The bottom surface **923** further defines a light inlet slot **934**. The light inlet slot **934** allows for a light signal to enter the cartridge system **900**. Particularly, the light inlet slot **934** allows for a light signal to enter the chip **936** and for the light signal to move through any waveguides in the chip **936** while interacting with analytes in the at least one detection microchannel **994** (see FIG. **9**F) before the light signal is deflected by one or more gratings (not shown) down to the detector unit **406** (see FIG. **6**).

(158) FIG. **9**C illustrates a view of the back surface **940** of one embodiment of a multiple-use cartridge system **900**. The housing includes a top portion **904** that is optionally removable from a bottom portion **910**. The male keying portions (**920**, **922**) are shown extending from the bottom portion **910** of the cartridge housing **902**.

(159) FIG. **9**D illustrates a view of the front surface **950** of one embodiment of a multiple-use cartridge system **900**. FIG. **9**E illustrates view of one side surface **960** of one embodiment of a single-use cartridge system **900**, the opposite side being a mirror image.

(160) FIG. **9**F illustrates a cross-section view downward of a multiple-use cartridge system **900** along the horizontal line of FIG. **9**E. The cartridge system **900** a storage means **907** as provided herein positioned within the cartridge housing **902**. The multiple-use cartridge system **900** includes a microfluidic system **970** for communicating or otherwise providing a means for a test sample composition to move through the cartridge system **900** and allow for detection and analysis of one or more analytes. An ingress port **972** is located on a front surface **950** (see FIG. **9**D) of the multiple-use cartridge system **900**. The ingress port **972** is in communication with a first microchannel section **974** having a first end **976** attached in communication with an ingress port check valve **973** and a second end **978** in communication with second microchannel section **979**. A filter **977** may be located anywhere within the first microchannel section **974**. A sample electrode **980** and reference electrode **982** are in contact with the second microchannel section **979**. Impedance may be measured between the sample electrode **980** and reference electrode **982** to confirm the presence of test sample composition.

(161) A valve test structure connection **984** is in communication with any test sample composition in the microfluidic system **970**. The valve test structure connection **984** may be fabricated from nitinol shape memory alloy and aids in the movement of test sample composition into the cartridge system **900**.

(162) The second microchannel section **979** includes a first end **988** in communication the first microchannel section **974** and a second end **990** in communication with a flow cell **992** having at least one detection microchannel **994**. The cartridge system **900** includes a detection region **993** that accommodates or is otherwise adapted to receive the chip **936** and flow cell **992**. The chip **936** is substantially transparent and allows the light signal to enter, interact with one or more waveguides channels (not shown; see e.g., part **316** of FIG. **3**C) and allow for binding of analyte flowing within the at least one detection microchannel **994** within the flow cell **992**.

(163) The detection microchannel **994** is in communication with a first end **996** of a third microchannel section **998**. The third microchannel section **998** includes a flow electrode **1000** to

approximate flow rate and is correlated with measured impedance. The third microchannel section **998** includes a second end **1002** in communication with the first end **1004** of a fourth microchannel **1006**. The fourth microchannel **1006** includes a second end **1008** in communication with a check valve **1010** which, in turn, is in communication with an egress port **1012**. The sample electrode **980**, reference electrode **982**, and flow electrode **1000** are each fabricated from inert nitinol or other corrosion-resistant conductive material.
(164) The multiple-use cartridge system **900** may further include a transmission component **1014** as provided herein. The multiple-use cartridge system **900** may further include a location means **1016** as provided herein.
(165) An exemplary embodiment of an alternative single-use cartridge system **1100** is illustrated in FIG. **10**. According to the illustrated embodiment, the cartridge system **1100** includes a connection mechanism **1102** (or snap-in rod) having opposing ends (**1104**, **1106**) extending from the housing **1108**. The connection mechanism **1102** aids in securing and interfacing the cartridge system **1100** with an interferometric system. Rising from the housing **1108**, are an injection ports **1110** A-D and outlet ports **1120** A-D. The injection ports **1110** A-D may be utilized for introducing a test sample, buffer or a test sample composition. The cartridge system includes four independent detection microchannel ports that are independently in communication with a corresponding detection microchannel (not shown) within a flow cell (not shown). Buffer may be pre-loaded in the flow cell. Any test sample composition waste may be collected from the outlet ports **1120** A-D.
(166) Healthcare Applications
(167) The interferometric systems provided herein may be utilized as a point of care system. The point of care testing may be carried out at or near the site of a where a healthcare target sample is obtained. In a healthcare setting, a medical professional may receive results in an efficient manner and any care decisions may be implemented immediately. By being mobile and utilized at the point of care, the systems provided herein provide a major technical advancement in the fight to diagnose and track pathogens that may give rise to global pandemics or be the cause of other rising, recurring or endemic diseases.
(168) The interferometric systems provided herein may be utilized to analyze and detect analytes taken from a bodily fluid or gaseous emission of the body. Such bodily fluids include, but are not limited to, blood, urine and saliva.
(169) The interferometric systems provided herein are suited to detect and quantify analytes such as one or more of a virus, bacteria, or small molecule such as a drug or drug metabolite. The interferometric systems provided herein are particularly suited to detect and quantify analytes of particular medical interest such as an illegal/illicit drug, SARS-CoV-2, *Yersinia pestis* (Plague), mycobacterium, influenza virus, hCG, human immunodeficiency virus, a particular vitamin, genetic mutation, IgG, IgE, and CD4/T-Cells. The interferometric systems provided herein are readily adaptable to new analytes within the healthcare environment as they emerge. The interferometric systems put high quality diagnostic results in the hands of healthcare professionals in an efficient manner.
(170) According to one particular embodiment, the interferometric system provided here may be utilized in connection with or otherwise equipped to a mobile vehicle. Suitable mobile vehicles include, but are not limited to, unmanned aerial vehicles (UAV), unmanned ground vehicles (UGV), drones, manned aircraft, and manned vehicles.
(171) Methods of Detection and Quantification
(172) FIG. **11** illustrates a method **1200** of detecting and quantifying the level of analyte in a healthcare test sample composition. The method includes the step of collecting **1202** or otherwise obtaining a target sample having one or more analytes. In different embodiments, the target sample may be taken from the appropriate target depending on the location and environment.
(173) According to one embodiment, the method further includes the optional step of entering **1204** a user identifier (ID) in the system. Additionally, an identification number associated with the

sample, analyte or interest or a combination thereof may be entered. The cartridge system utilized may be equipped with a label or sticker carrying identifying such information. The label or sticker may include a QR code including such information. The label or sticker may be removed prior to use. Identifying information may include metadata such as time, GPS data, or other data generated by the interferometric system described herein.

(174) According to one embodiment, the method further includes the step of introducing the target sample to the interferometric system **1206**. According to one embodiment, target sample is introduced to the cartridge by a separate device such as a syringe or pump. According to one embodiment, target sample is introduced by an injection device. According to one embodiment, the injection device may be permanently attached to the cartridge system. According to one embodiment, the injection device is a pipette. According to one embodiment, the injection device is a syringe. According to one embodiment, the injection device is a lance, pipette or capillary tube. When utilizing a multiple-use cartridge system, the cartridge system may be fitted to a tube or other transfer mechanism to allow the sample to be continuously taken from a large amount of fluid that is being monitored.

(175) According to one embodiment, the method further includes the step of mixing **1208** the target sample with a buffer solution to form a test sample composition. In a multiple-use cartridge system, such a step may occur prior to the test sample composition being introduced to the cartridge system. In a single-use cartridge system, such a step may occur in the mixing bladder with the assistance of a pump.

(176) The method of detecting and quantifying the level of analyte in a sample includes initiating waveguide interferometry **1210** on the test sample composition. Such a step may include initiating movement of the light signal through the cartridge system as provided herein and receiving the light signal within the detector unit. Any changes in an interference pattern are representative of analyte in the test sample composition. Particularly, such changes in an interference pattern generate data related to one or more analyte in the test sample composition. According to one embodiment, the step of initiating **1210** waveguide interferometry on the test sample composition includes the step of correlating data from the phase shift with calibration data to obtain data related to analyte identity, analyte concentration, or a combination thereof.

(177) According to one embodiment, the method further includes the step of processing **1212** any data resulting from changes in the interference pattern. Such changes in interference pattern may be processed and otherwise translated to indicate the presence and amount of an analyte in a test sample composition. Processing may be assisted by software, processing units, processor, servers, or other component suitable for processing. The step of processing data may further include storing such data in storage means as provided herein.

(178) According to one embodiment, the method further includes the step of transmitting a data signal **1214**. The signal may result in the display of data on the system. The step of transmitting data may include displaying the analyte levels via projecting any real time data on a screen as described herein. The step of transmitting data may include transmitting any obtained data to a mobile phone, smart phone, tablet, computer, laptop, watch or other wireless device. The data may also be sent to a device at a remote destination. The remote destination device may be a locally operated mobile or portable device, such as a smart phone, tablet device, pad, or laptop computer. The destination may also be smart phone, pad, computer, cloud device, or server. In other embodiments, the remote destination may be a stand-alone or networked computer, cloud device, or server accessible via a local portable device. A diagnosis of an infection in a healthcare environment may be based on the analyte quantity. The diagnosis may be based on the use of one or more immunoglobulins as detection materials.

(179) The method may optionally include the step of disposing of the test sample composition **1216** per legal requirements. Such legal requirements assure that any sample still containing unacceptable levels of pathological contamination are disposed of properly so as not to cause harm

to a user or the environment.
(180) According to one embodiment, the method further includes the step of initiating **1218** a cleaning or remedial countermeasure against any analyte detected. Such remedial measure may include introducing cleaning chemicals or beneficial microorganisms to the healthcare environment. The remedial measures may work to kill or otherwise neutralize any unwanted analyte present in the healthcare environment where a sample was taken.
(181) Although the present specification describes components and functions that may be implemented in particular embodiments with reference to particular standards and protocols, the invention is not limited to such standards and protocols. For example, standards for Internet and other packet switched network transmission (e.g., TCP/IP, UDP/IP, HTML, HTTP) represent examples of the state of the art. Such standards are periodically superseded by faster or more efficient equivalents having essentially the same functions. Accordingly, replacement standards and protocols having the same or similar functions as those disclosed herein are considered equivalents thereof.
(182) Although specific embodiments of the present invention are herein illustrated and described in detail, the invention is not limited thereto. The above detailed descriptions are provided as exemplary of the present invention and should not be construed as constituting any limitation of the invention. Modifications will be apparent to those skilled in the art, and all modifications that do not depart from the spirit of the invention are intended to be included with the scope of the appended claims.
PROPHETIC EXAMPLE 1
SARS-CoV-2 Detection and Quantification
(183) A healthcare setting or mobile testing unit may be set up to aid in high throughput detection and quantification of SARS-CoV-2 in a patient. A medical professional or other trained user may obtain a sample. A user may obtain a sample in form of a small blood sample. In the case of a blood sample, the user may clean a patient's finger with an alcohol swab. The user may prick or otherwise lance the finger and obtain an effective amount (aliquot) of blood for the system. The aliquot of blood may be obtained via a medical device such as a disposable hemoglobin sterile pipette. Since the systems provided herein only need a small amount of sample, the system eliminates the need for a large-volume sample and centrifuging thereby significantly simplifying use and speeding up the process compared to existing technologies that require large sample volume.
(184) A cartridge is then placed inside the system's detector component, if not already present. The cartridge may be fully replaceable and disposable after each use. The user may optionally then enter a user identifier (ID) in the system and the system optionally transmits that information to the remote server for authentication or stores the information locally. Any of a number of identifier labelling techniques, such as radio frequency identifiers (RFIDs) on or within a sample may be used. Alternatively, a unique serial number, code or other identifier associated with a sample may be manually entered into the system and optionally transmitted to a remote server. Additionally, the user may use the system to scan in or manually enter one or more substance/contaminant identifiers, such as a Universal Product Code (UPC) for the one or more analytes believed to be present in the sample and to inform the remote server of the one or more analytes.
(185) The system may also include geolocation information in its communications with the server, either from a GPS sensor included in the system or a GPS software function capable of generating the location of the system in cooperation with a cellular or other communication network in communication with the system.
(186) For detection and quantification of SARS-CoV-2, one or more antibodies specific to the surface proteins on SARS-CoV-2 will be included on the sensitive layer. The antibodies may be specific to spike, envelope, membrane and nucleocapsid proteins found on the surface of SARS-CoV-2.
(187) If present, the user may initialize the system by pressing a start button or other similar means

to initiate any electronic components present in the system. If present, the user may optionally press an injection bulb or similar mechanical component to inject a buffer solution into the cartridge. Any display on the system may then provide visual signal that the system is ready for sample introduction (e.g., signal "READY"). If present, the user may then press another external display or button to signal the system that a sample is ready to be introduced (e.g., "SAMPLE").

(188) To introduce a sample, an aliquot of the sample may be added through a sample collecting component. Such a step may be accomplished via a disposable pipette or similar device that is suited for storing a sample until needed. Next, a user may press the injection bulb or similar mechanical component to mix the aliquot of sample and buffer and introduce the mixture to the cartridge. According to an alternative embodiment, the buffer may be mixed with the aliquot sample in a separate step prior to introduction to the cartridge. A user may then press the injection bulb or similar mechanical component a one or more times to ensure the sample mixture has fully transitioned or otherwise migrated to the cartridge and begins flowing across the waveguide channels on the waveguide. Upon arrival at the waveguide, detection and quantification processes are undertaken. Depending on the goal of the point of care procedure, one or more biological components in the sample will bind to the sensitive layer of the waveguide channels thereby altering the evanescent field above the waveguide channels. The changes in the interference pattern will then create an electronic signal that can be translated to produce a reading on a display on an external surface of the system. Any medical devices used during use may then be disposed of properly in a biological waste container. The cartridge within the system may then be removed and replaced with a new cartridge or cleaned prior to next use. The cartridge may be fully disposable and placed in a biological waste container along with the medical devices utilized during use.

PROPHETIC EXAMPLE 2

Avian Influenza Detection

(189) A healthcare setting or mobile testing unit may be set up to aid in high throughput detection and quantification of avian influenza in a patient. A medical professional or other trained user may carry out the same general steps as set forth in Prophetic Example 1. When detecting and quantifying avian influenza in a sample, any avian influenza analyte particles may be captured by utilizing avian influenza specific antibodies on the sensitive layer that are configured to bind to avian influenza surface protein. In some embodiments, monoclonal and polyclonal antibodies may be utilized on the sensitive layer. In other embodiments, polyclonal antibodies may be utilized to attach to one or more antigenic sites on the viral protein thereby reducing the likelihood of a false negative indication (avoid looking at only a single antigenic epitope).

PROPHETIC EXAMPLE 3

Drug Detection

(190) The systems provided herein may be utilized to aid in high throughput detection and quantification of a small molecule such as a drug or drug metabolite in a patient. Such detection methods may be particularly useful for job screening or for legal reasons. Such detection methods may also be particularly useful in emergency departments of healthcare facilities where a drug overdose is suspected. A medical professional or other trained user may carry out the same general steps as set forth in Prophetic Example 1. When detecting and quantifying drug and drug metabolites in a sample, any target small molecule may be captured by utilizing molecularly imprinted polymers on the sensitive layer that are configured to bind to small molecules. Data regarding a specific class of drug or specific drug compound may be provided in an efficient manner to a user. Any corresponding report may be submitted wirelessly to a third party such as a prospective employer or law enforcement agency, if needed.

PROPHETIC EXAMPLE 4

Pathogenic Detection and Quantification in Dental Offices

(191) The systems provided herein may be utilized to aid in high throughput detection and quantification of SARS-CoV-2 via implementation of systems in dental offices. Dental offices deal

with several oral diseases but are also subject to common disease and viral threats such as HIV, Hepatitis, Flu, and Corona Viruses such as SARS-CoV-2. The United States dental industry has over 150,000 dental hygienists which see roughly 8 patients a day or roughly 1,200,000 per day nationwide (0.38% of United States population daily or approximately 7.5% of population monthly). More than half of the United States population visits a dental hygienist at least once per year. Dental hygienists are trained to deal with both saliva and blood and the real potential that the patient could be contagious with various analytes such as SARS-CoV-2. Using the systems provided herein to detect for these potential viral analytes prior to a dental exam can serve at least two purposes: The systems provided herein can diagnose a patient while providing early intervention as well as monitor pathogens to help prevent outbreak, epidemic, or pandemic.

(192) According to one embodiment, the systems provided herein may be utilized to screen or otherwise detect a pathogen for each patient prior to or upon entering a dental office (HIPAA compliance required). The system may be located in a lobby or separate area such that results regarding pathogenic infection may be provided prior to entry into the office and subsequent dental treatment. Screening may occur with a saliva or blood sample from a patient. Such a screening process may be financially subsidized by a patient's dental insurance as well as supported by both the ADA and the AMA. According to one embodiment, the systems provided herein may be utilized to provide the dental office with an additional source of revenue via patient screening.

(193) FIG. **12**A illustrates a quantification and monitoring system for analytes within an aqueous target sample from a rinse sink. As illustrated in FIG. **12**A, an interferometric system **1300** may be utilized to monitor the rinse water **1310** flowing out of a dental “rinse” sink **1320**. In use, the rinse water **1310** may move through a first drain pipe **1330** and diverted by a valve **1340** between a collection pipe **1350** for the interferometric system **1300** and a sewer/septic pipe **1360**.

(194) FIG. **12**A illustrates a quantification and monitoring system for analytes within an aqueous target sample from suction line. As illustrated in FIG. **12**B, an interferometric system **1400** may be utilized to monitor the rinse water **1410** flowing through a dental suction line **1420** used during a dental cleaning or other procedure. In use, the rinse water **1410** may move through a first drain pipe **1430** and diverted by a valve **1440** between a collection pipe **1450** for the interferometric system **1400** and a sewer/septic pipe **1460**.

(195) The results of monitoring in a dental facility may be sent to a third monitoring service. According to such an embodiment, the interferometric system provides a reliable sampling of the general United States population. By enumerating the patients, geolocating the suction line or sink, and sending the data to a central location (e.g., cloud-based server), the system may function as a digitized monitoring system for mapping results across the United States. After sampling rinse water from a particular patient, the system may initiate a cleaning and decontamination step of the suction line, pipes and any components within the system. According to a particular embodiment, the cartridge system within the interferometric system may be removed and replaced with a new cartridge after each use or cleaned prior to next use. According to a particular embodiment, the cartridge system within the interferometric system may be removed and replaced with a new cartridge periodically.

(196) The interferometric system can also become part of the normal maintenance of those managing the dental office making the detection and monitoring methods seamless. The statistical relevance of this type of monitoring allows for non-HIPAA collection of data while monitoring the health of a particular region and, in turn, the overall United States. In the event HIPAA laws require authorization for this type monitoring, a bypass switch can be installed and the system will reduce the sample size for analysis by that number.

## Claims

1. A portable interferometric system for detection and quantification of analyte within a healthcare test sample composition, the system comprising: an optical assembly unit, the optical assembly unit comprising a light unit and a detector unit each adapted to fit within a portable housing unit; a cartridge system adapted to be inserted in the portable housing unit and removed after one or more uses, the cartridge system comprising: an interferometric chip including one or more waveguide channels having a sensing layer thereon, the sensing layer adapted to selectively bind or otherwise be selectively disturbed by one or more analytes within the healthcare test sample composition; and a flow cell wafer; and a cartridge housing enclosing the interferometric chip and flow cell wafer; and an alignment means for aligning the cartridge system within a cartridge recess in the optical assembly unit upon insertion thereby providing optical and microfluidic alignment of the interferometric chip and flow cell wafer.
2. The portable interferometric system of claim 1, wherein the portable housing is sized and shaped to fit in a user's hand.
3. The portable interferometric system of claim 1, further comprising at least one display unit.
4. The portable interferometric system of claim 1, further comprising a camera, the camera adapted to capture a photo or video.
5. The portable interferometric system of claim 1, wherein the sensing layer comprises one or more antigens, antibodies, DNA, aptamers, polypeptides, nucleic acids, carbohydrates, lipids, or molecularly imprinted polymers, or immunoglobulins suitable for binding one or more analytes within a healthcare test sample composition.
6. The portable interferometric system of claim 1, configured to analyze the light signals from two or more waveguide channels to detect the presence of an analyte that individual waveguides could not have detected alone.
7. The portable interferometric system of claim 1, wherein the one or more waveguide channels each comprises a sensitive layer and wherein each sensitive layer may detect a different analyte.
8. The portable interferometric system of claim 7, wherein the sensitive layer is configured to bind one or more small molecules, antibodies, virus antigens, virus proteins, bacteria, fungi, pathogen, RNA, chemical, mRNA or any combination thereof.
9. The portable interferometric system of claim 1, having an analyte detection limit down to about 1.0 picogram/L.
10. The portable interferometric system of claim 1, having an analyte detection limit down to about 1000 pfu/ml.
11. The portable interferometric system of claim 1, wherein the detector unit has sensitivity to at least 2 pixels per diffraction line pair.
12. The portable interferometric system of claim 1, further comprising a location means adapted to determine a physical location of the system.
13. The portable interferometric system of claim 1, wherein the analyte is a pathogen.
14. A method of detecting and quantifying the level of analyte in a healthcare test sample composition, the method comprising the steps of: collecting a healthcare target sample containing one or more analytes; optionally entering an identification associated with the healthcare target sample; introducing the healthcare target sample to the portable interferometric system of claim 1; optionally, mixing the healthcare target sample with a buffer solution to form a healthcare test sample composition; initiating waveguide interferometry on the healthcare test sample composition; processing any data resulting from the waveguide interferometry; and optionally, transmitting any data resulting from the waveguide interferometry.
15. The method of claim 14, wherein the step of transmitting data includes wirelessly transmitting analyte detection and quantification data to a mobile device or server.
16. The method of claim 14, further comprising the step of displaying data related to the presence of analyte in the test sample composition on a display unit.

17. The method of claim 14, wherein the healthcare target sample is taken from water, soil, air, exhaled breath, skin, hair, or a bodily fluid or gaseous emission of the body.
18. The method of claim 17, wherein the healthcare target sample is in the form of, dissolved in, or suspended in a liquid or a gas.
19. The method of claim 14, wherein the data resulting from the waveguide interferometry is provided at or under 30 minutes.
20. The portable interferometric system of claim 1, wherein the cartridge housing comprises a vent port for allowing air to exit and prevent bubble formation.